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**What is the problem you are solving? (50 words max)**

we're building an AI-powered healthcare assistant for rural India that works via voice calls on feature phones and through a lightweight app on smartphones. It helps ASHA workers detect high-risk pregnancies, prevent anemia, access offline support, and spread health scheme awareness in local languages with risk free Loan Assessment.

We will use **TensorFlow Lite** and **Keras** to build **lightweight NLP models** for voice understanding and **rule-based ML models** for health risk prediction, optimized for **on-device inference** to ensure offline access and performance on low-end Android devices.

**Making an  
Responsible AI**

**Describe your solution. How different is it from any of the other existing ideas? How will it be able to solve the problem? USP of the proposed solution? What is the intended impact of your solution (max 350 words).**

**Aarogya Saathi** is an **AI-powered healthcare assistant** built for **rural India**, aiming to bridge the healthcare gap through accessible technology. The solution works via **voice calls on basic feature phones** and a **lightweight Android app**, both supporting **multiple Indian languages**. It empowers **ASHA workers** and **rural citizens** to access **reliable health support** without requiring **internet connectivity or smartphones**, **btw** in India people prefer there cultural food over balanced diet fixing there thought process with better knowledge in there own local language.

#### **Main Features:**

- It uses **lightweight on-device AI models** and **IVR (Interactive Voice Response)** to deliver **voice-guided pregnancy risk assessments**, even in **low-network areas**.
- The system can understand **local-language medical queries** and provide **instant health guidance** using **NLP tailored for Indian dialects**.
- Aarogya Saathi is designed to be **offline-first**, with features like **smart caching of medical data and health schemes**, and **automatic syncing** once the internet is available.

#### **What Makes It Unique:**

- Unlike most digital health solutions that rely on **smartphones and stable internet**, this solution works on **any phone**.
- The approach is **voice-first, language-first**, and focuses on **inclusive design** for communities with **low digital literacy and to teach them also about balanced Diet**.

#### **Our Goal:**

The intended impact is to **reduce maternal and infant mortality**, provide **ASHA workers with intelligent tools**, and bring **equitable healthcare access** to India's most underserved regions using **scalable AI-driven tech** and Educate people about **balanced diet**.

## How is this solution scalable? (100 words max)

- ◆ Built with **modular, open-source AI tools** – easily customizable for different regions and languages
- ◆ **Offline-first architecture** – works without internet, ideal for rural deployment
- ◆ Compatible with **feature phones and low-end smartphones** – no need for expensive devices
- ◆ **Lightweight on-device models** with **cloud sync** – ensures smooth performance and scalability
- ◆ Can integrate with **existing government schemes and NGO networks** for wide rollout
- ◆ **Low cost, high reach** – scalable across India's diverse geographies with minimal infrastructure

Who is the primary user of your solution, and explain how your solution will leverage open-source AI to address the aspects mentioned in the [Key Design Guidelines](#) (max 200 words).

The **primary users** of Aarogya Saathi are people in **rural and underserved areas** with **limited access to healthcare and digital infrastructure**. While designed to support **ASHA workers**, with important features is meant for **anyone with a basic phone**, especially **pregnant women and families** seeking trusted medical guidance.

Aarogya Saathi leverages **open-source AI models** like **Whisper** for speech recognition and Indicbert for understanding regional languages. These models are optimized to run **offline** and on **low-end devices**, ensuring accessibility even without internet. The system uses **IVR** and **lightweight on-device AI** to offer voice-based health support in multiple Indian languages , even it can be used on shared device with security features , we will be making it keeping every aspect of society in mind.

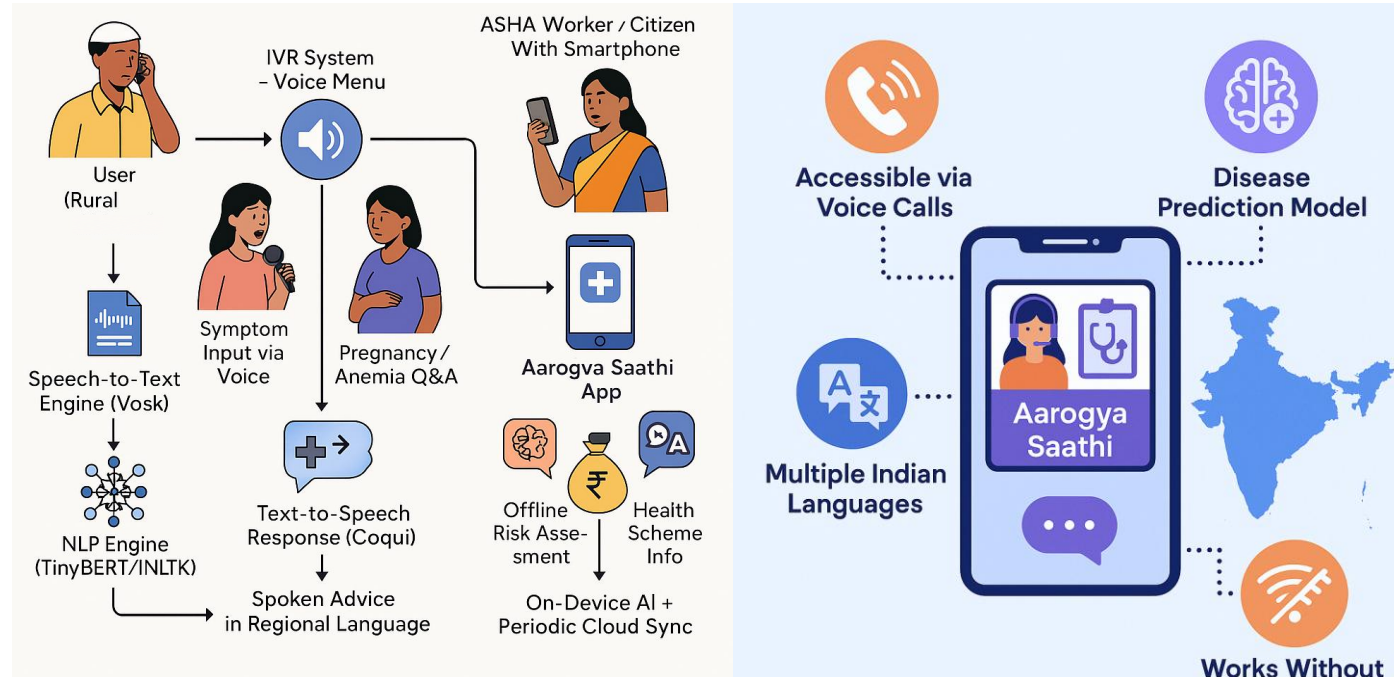
This approach supports the **Key Design Guidelines** by being:

- Inclusive** (supports all phones and local languages)
- Low-resource friendly** (works offline, minimal infrastructure needs)
- Ethical** (respects user privacy, open-source based)
- Scalable and affordable** (built using flexible, community-driven tools)

By combining **open-source AI** with a **voice-first, offline-first** design, Aarogya Saathi delivers critical health support to those who need it most.

## List of features offered by the solution

It is always better to add a few visual representations (drawings/sketches/illustrations etc.) to your presentation, it adds to the power through which it reaches the audience.



## What open-source AI tools and technologies will you use to design the solution? (Please list all.)

| Category                          | Tool/Technology                             |
|-----------------------------------|---------------------------------------------|
| Voice & IVR System                | Asterisk, Twilio, Google Dialogflow CX      |
| Speech Recognition                | Vosk, Mozilla DeepSpeech                    |
| Text-to-Speech (TTS)              | Coqui TTS, Google TTS API                   |
| Natural Language Processing (NLP) | BERT (TinyBERT), IndicNLP, spaCy            |
| Lightweight AI Model Framework    | ONNX, TensorFlow Lite, PyTorch Mobile       |
| Mobile App Development            | Android Studio, Kotlin, Java                |
| Backend & APIs                    | FastAPI, Node.js                            |
| Database & Storage                | Firebase, PostgreSQL, SQLite                |
| Cloud Infrastructure              | AWS (Lambda, EC2, S3, DynamoDB), CloudWatch |
| Language Support                  | iNLTK, AI4Bharat models                     |
| Data Sync & Offline Access        | WorkManager (Android), Room DB, Retrofit    |
| Security & Privacy                | JWT, HTTPS, OAuth 2.0, local encryption     |

We use a range of open-source technologies including **Asterisk, Vosk, Mozilla DeepSpeech, Coqui TTS, TinyBERT, IndicNLP, spaCy, ONNX, TensorFlow Lite, PyTorch Mobile, Android Studio, Kotlin, Java, FastAPI, Node.js, PostgreSQL, SQLite, Room DB, Retrofit, WorkManager, iNLTK, AI4Bharat models, JWT, OAuth 2.0, and local encryption** methods. These tools ensure our solution remains cost-effective, adaptable, and accessible for deployment and scaling across rural India.

**Why are these open-source technologies the most appropriate for your solution?**  
**(150 words max)**

These open-source technologies are ideal for our solution because they offer **cost-effectiveness**, **customizability**, and **offline capabilities**, which are essential for deployment in **rural India**. Tools like **Vosk**, **Coqui TTS**, and **TinyBERT** are lightweight and support regional languages, enabling AI functionalities on low-end devices and basic feature phones. Frameworks such as **TensorFlow Lite** and **PyTorch Mobile** allow efficient on-device AI processing, reducing dependency on internet connectivity.

Open-source platforms like **FastAPI**, **Node.js**, and **SQLite** ensure fast, scalable backend operations with minimal infrastructure. Using **Room DB**, **WorkManager**, and **Retrofit** allows reliable offline-first mobile experiences. Moreover, tools like **iNLTK** and **AI4Bharat** models provide localized language support, enhancing usability for native speakers. Open-source standards like **JWT** and **OAuth 2.0** maintain strong security and data privacy. Overall, this tech stack ensures our solution remains inclusive, flexible, and scalable in resource-constrained environments.

## Describe the Solutions Architecture (500 words)

The *Aarogya Saathi* architecture is built to function in low-connectivity rural areas using a hybrid model of feature phones (via IVRS) and lightweight Android apps. It combines open-source AI tools with scalable cloud infrastructure to offer offline-first healthcare support, speech interaction, and multilingual access.

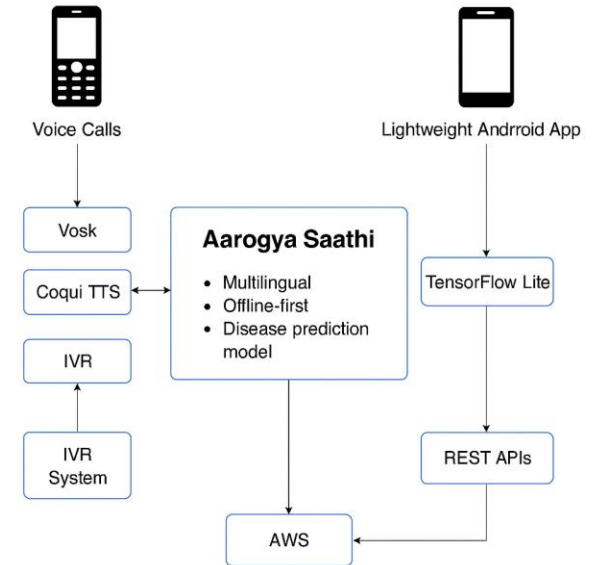
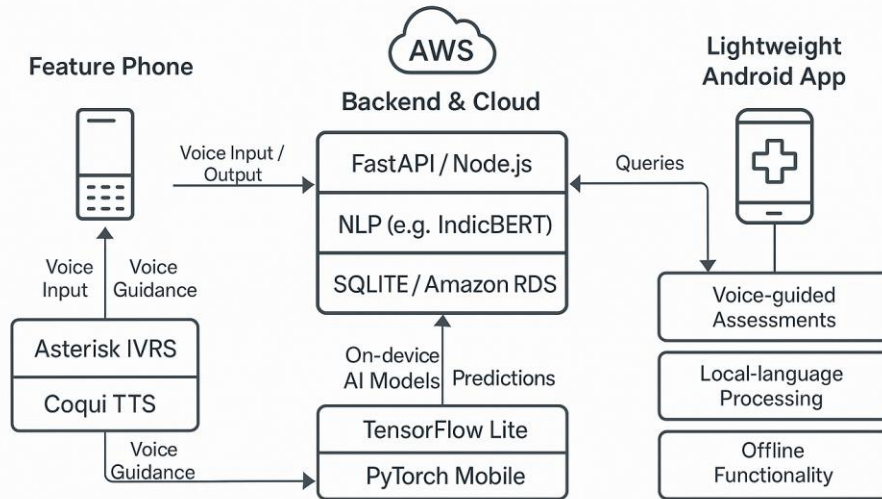
### Key Architectural Highlight:

- **On-device AI processing** with **TensorFlow Lite** and **PyTorch Mobile** for offline risk prediction and symptom analysis.
- **Voice input/output pipeline** using **Vosk** (ASR) and **Coqui TTS**, enabling interaction in local languages without internet.
- **IVRS system** built on **Asterisk**, hosted on **AWS EC2**, supports voice navigation on feature phones.
- **Multilingual NLP** using **iNLTK**, **IndicBERT**, and **AI4Bharat** for understanding and responding to health queries in Indian languages.
- **Mobile app** developed in **Kotlin** with offline capability via **Room DB**, background sync using **WorkManager** and **Retrofit**.

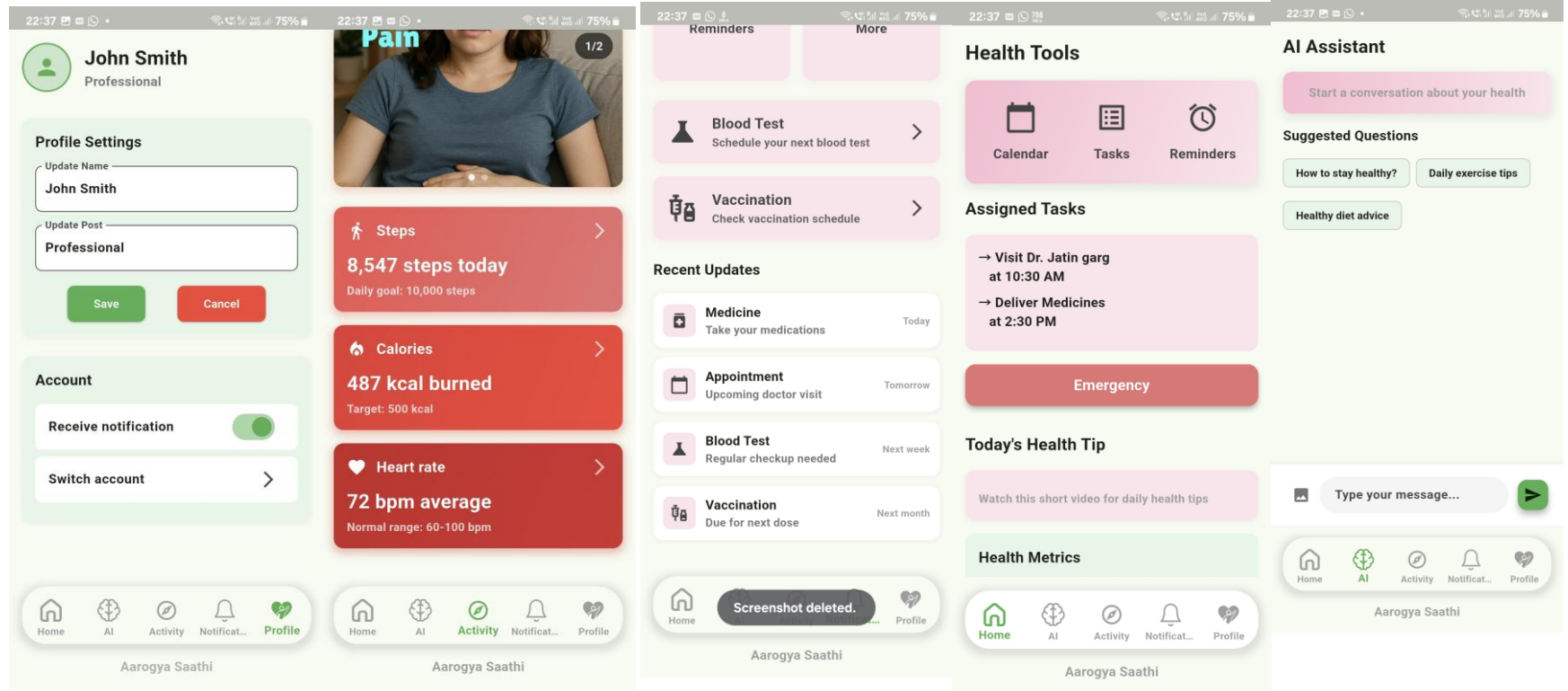
- **REST APIs** built with **FastAPI** and **Node.js**, managing secure communication between app/IVRS and backend.
- **Backend services** hosted on **AWS**, utilizing **Amazon S3** (storage), **Amazon RDS/SQLite** (database), and **AWS Lambda** (event processing).
- **UI/UX** designed using **Jetpack Compose**, focused on accessibility and ease of use.
- **Security** ensured with **JWT** and **OAuth 2.0** for secure user identification and role-based access.
- **Monitoring** through **Prometheus** and **Grafana** to track system health and performance.
- **Containerized deployment** using **Docker** and **AWS ECS** for easy scalability.
- **Model improvement loop** using anonymized user data and retraining on **Hugging Face**, **scikit-learn**, and **pandas**.

Provide a high-level architecture diagram or a use-case diagram of your proposed solution

## Solution Architecture



# Snapshots of prototype



**What datasets will your solution use? Are they publicly available, synthetic, or user-generated?**

**Yes , Most of them are publicly available.**

**•HMIS (Health Management Information System)**

- Public maternal health data from MoHFW India
- [hmis.nhp.gov.in](http://hmis.nhp.gov.in)

**•RCH Portal Data**

- Reproductive & Child Health statistics
- [rch.nhm.gov.in](http://rch.nhm.gov.in)

**•AI4Bharat Datasets**

- Multilingual speech/text for NLP in Indian languages
- [ai4bharat.iitm.ac.in](http://ai4bharat.iitm.ac.in)

**•Indic NLP / iNLTK**

- Indian language corpus for voice/text models

**•Synthetic Data**

- Simulated user conversations for model training

**•User-Generated Data**

- Real app/IVRS inputs (anonymized, with consent)

**Does your solution require cloud-based computation, or can it work with on-device processing? If cloud-based, how do you plan to address connectivity challenges and cost constraints?**

Yes, our solution follows a **hybrid architecture** — it supports **both on-device processing** and **cloud-based computation**, depending on the use case and device capabilities.

### **On-Device Processing**

- For basic diagnosis, symptom checks, and pregnancy risk assessment, we use **lightweight AI models (e.g., TensorFlow Lite, TinyBERT)** that run directly on low-end Android smartphones without requiring internet.
- This ensures **offline functionality**, which is crucial for rural areas with poor or no connectivity.
- The app uses **local storage (Room DB)** and **WorkManager** for delayed data synchronization when connectivity is restored.

### **Cloud-Based Processing**

- Feature phone users connect via **IVR**, where **speech processing (Vosk/DeepSpeech)** and **NLP tasks** are handled in the cloud.
- Backend logic, language model handling, analytics, and voice synthesis are hosted on **AWS services** such as EC2, Lambda, RDS, and S3.

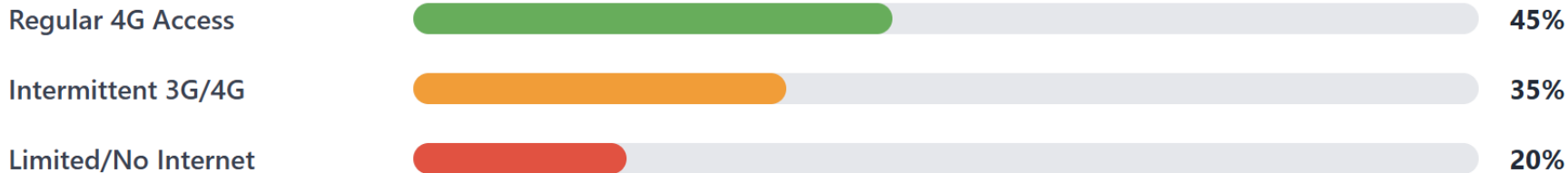
## Addressing Connectivity and Cost Constraints

- The solution uses **smart caching** and an **offline-first design** to reduce dependence on real-time connectivity.
  - Batch data synchronization** is performed only when the device regains connectivity, minimizing bandwidth usage.
  - Cost efficiency** is achieved by leveraging **serverless AWS Lambda functions, auto-scaling EC2 instances**, and **open-source AI tools**, eliminating the need for expensive third-party APIs.
- This balanced approach ensures the solution remains both **accessible** and **cost-effective**, regardless of the user's device or infrastructure limitations.

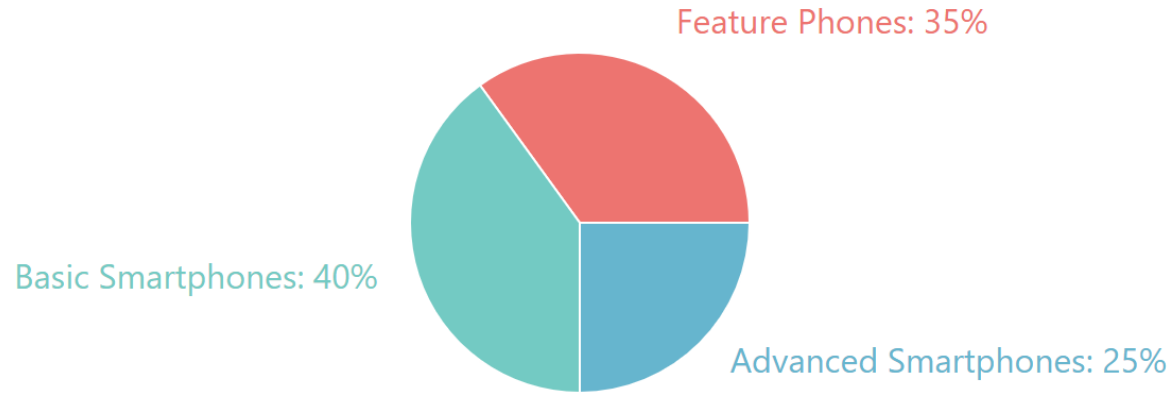
## Planned future enhancements or scalability considerations

- AI-based Snake Identification:** Integrate a trained model to instantly recognize snake species from uploaded images.
- Real-time Doctor Alerts:** Notify nearby doctors instantly for critical cases.
- Multilingual Support:** Expand accessibility for diverse rural users.
- Role-Based Dashboard Enhancements:** Custom views for ASHA workers, doctors, and administrators.
- Offline Mode with Sync:** Support for low-connectivity rural areas with automatic data sync when online.
- Scalable Backend:** Deploy on cloud platforms (AWS/GCP) with load balancing and containerization (Docker, Kubernetes) for future user growth.

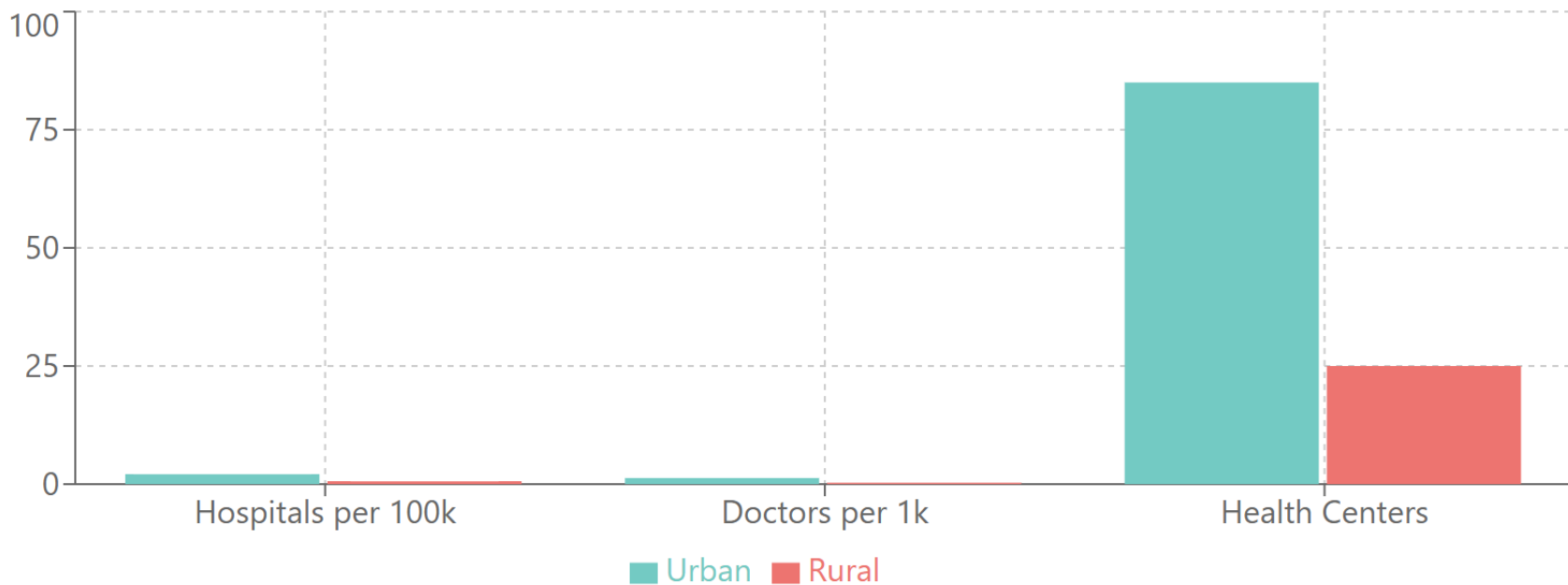
## Internet Connectivity in Rural Areas



## Device Usage in Rural India



## Urban vs Rural Healthcare Gap



## Maternal Mortality

**20%**

Projected reduction

## Early Detection

**35%**

Improvement achieved

## Healthcare Seeking

**55%**

Behavior improvement

## Preventive Care

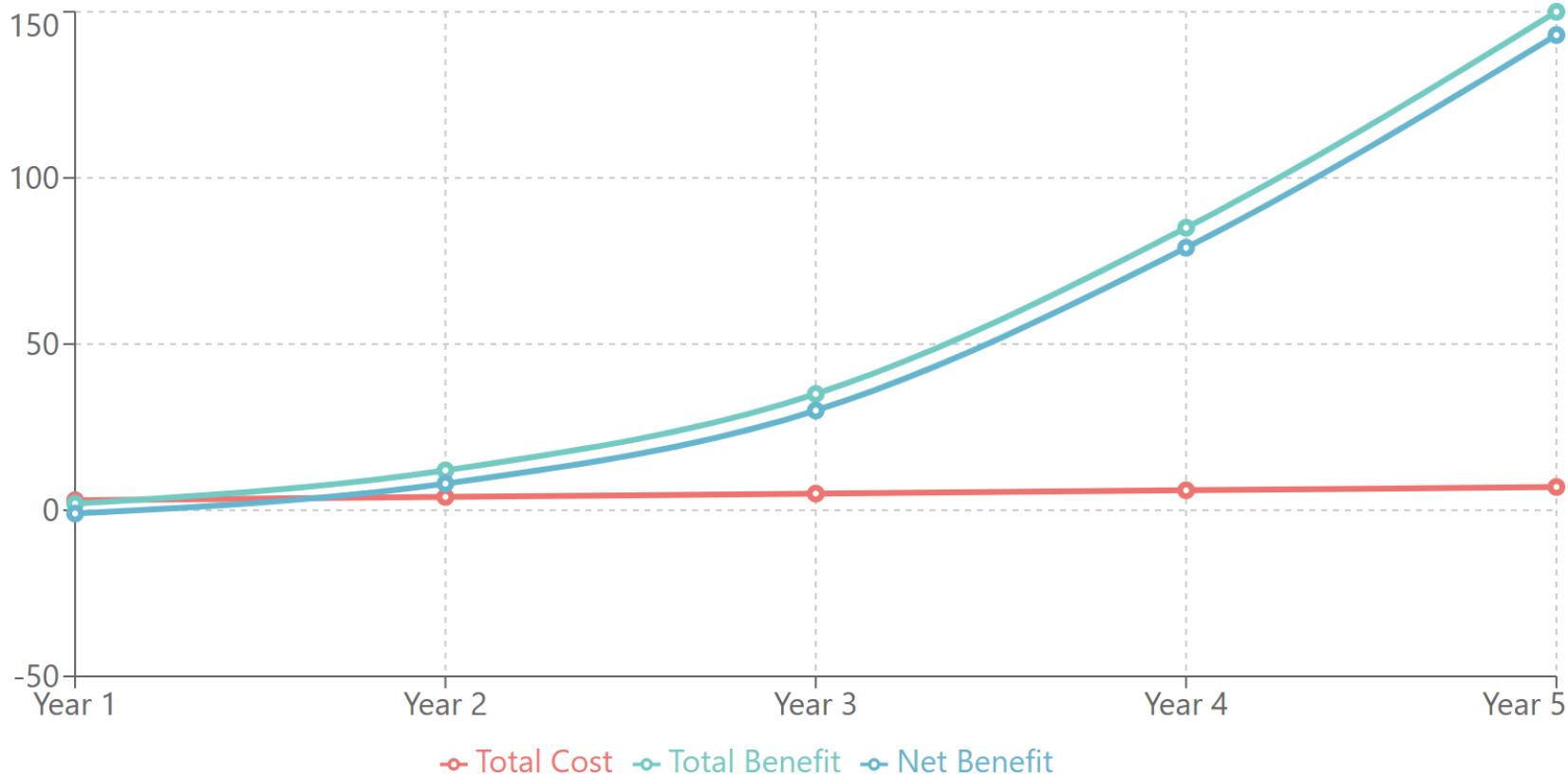
**40%**

Adoption increase

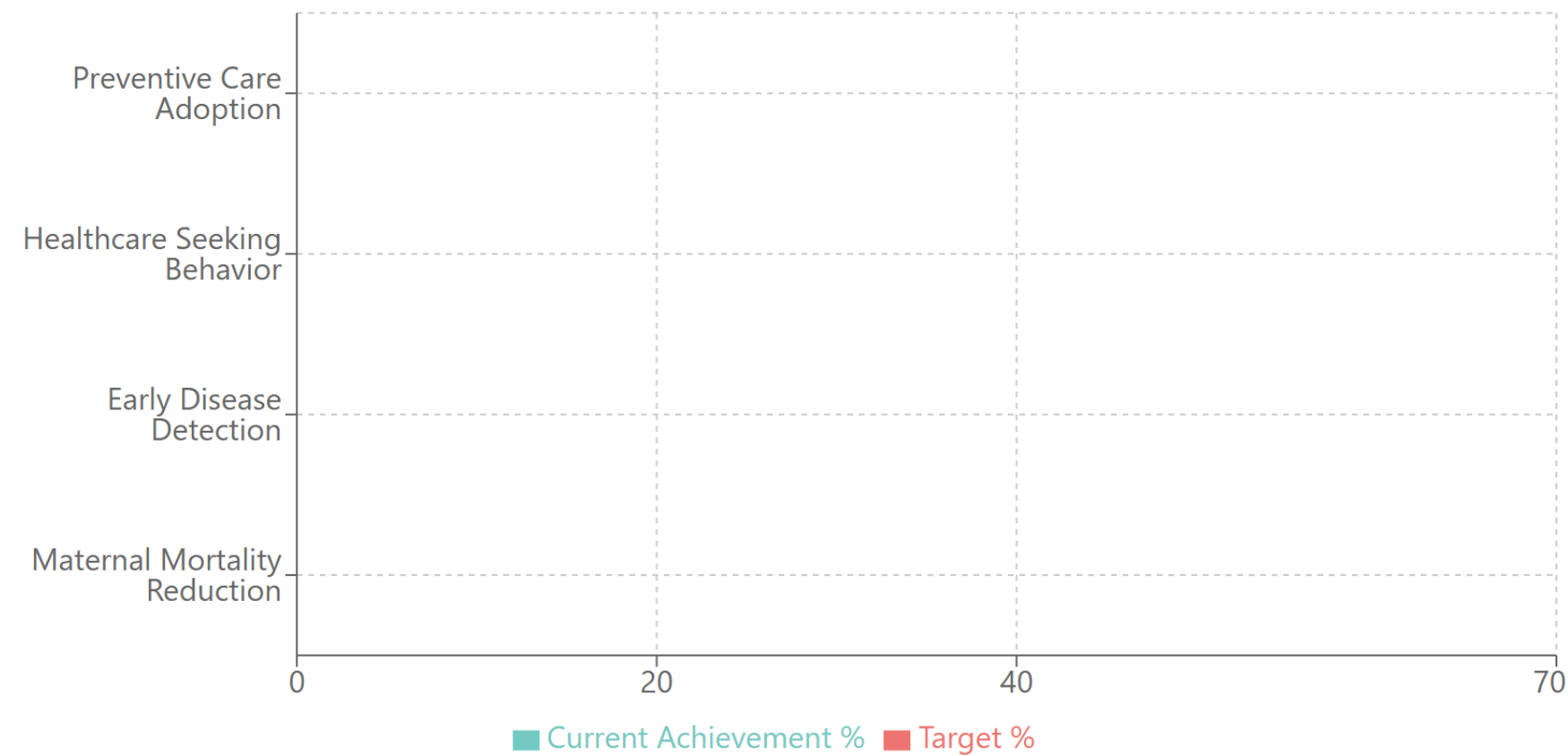
# Infrastructure Scaling Requirements

| USER SCALE | SERVER (VCPUS) | STORAGE | BANDWIDTH | MONTHLY COST |
|------------|----------------|---------|-----------|--------------|
| 100K Users | 50             | 10TB    | 1Gbps     | ₹5 Lakhs     |
| 1M Users   | 500            | 100TB   | 10Gbps    | ₹35 Lakhs    |
| 10M Users  | 5000           | 1PB     | 100Gbps   | ₹2.5 Crores  |

## Cost-Benefit Analysis (₹ Crores)



# Health Impact Metrics vs Targets



## High Risk Items

**2**

Requiring immediate attention

## Medium Risk Items

**3**

Monitoring required

## Overall Risk Score

**6.5/10**

Manageable risk level

## Mitigation Coverage

**85%**

Risks with mitigation plans

## Expected Feature Utilization

Health Information

85%



Pregnancy Tracking

60%



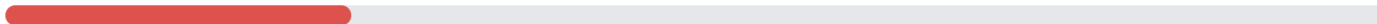
ASHA Dashboard

75%



Emergency Guidance

25%



## Technical Performance Targets

App Response Time

<2s offline



Data Sync Success

95%+



Offline Functionality

80%



# Risk Mitigation Strategies

## High Priority: Regulatory Compliance

Early engagement with regulatory bodies, legal compliance framework implementation

## High Priority: Content Accuracy

Medical expert validation, regular content updates, peer review process

## Medium Priority: Connectivity Issues

Robust offline capabilities, data compression, sync optimization

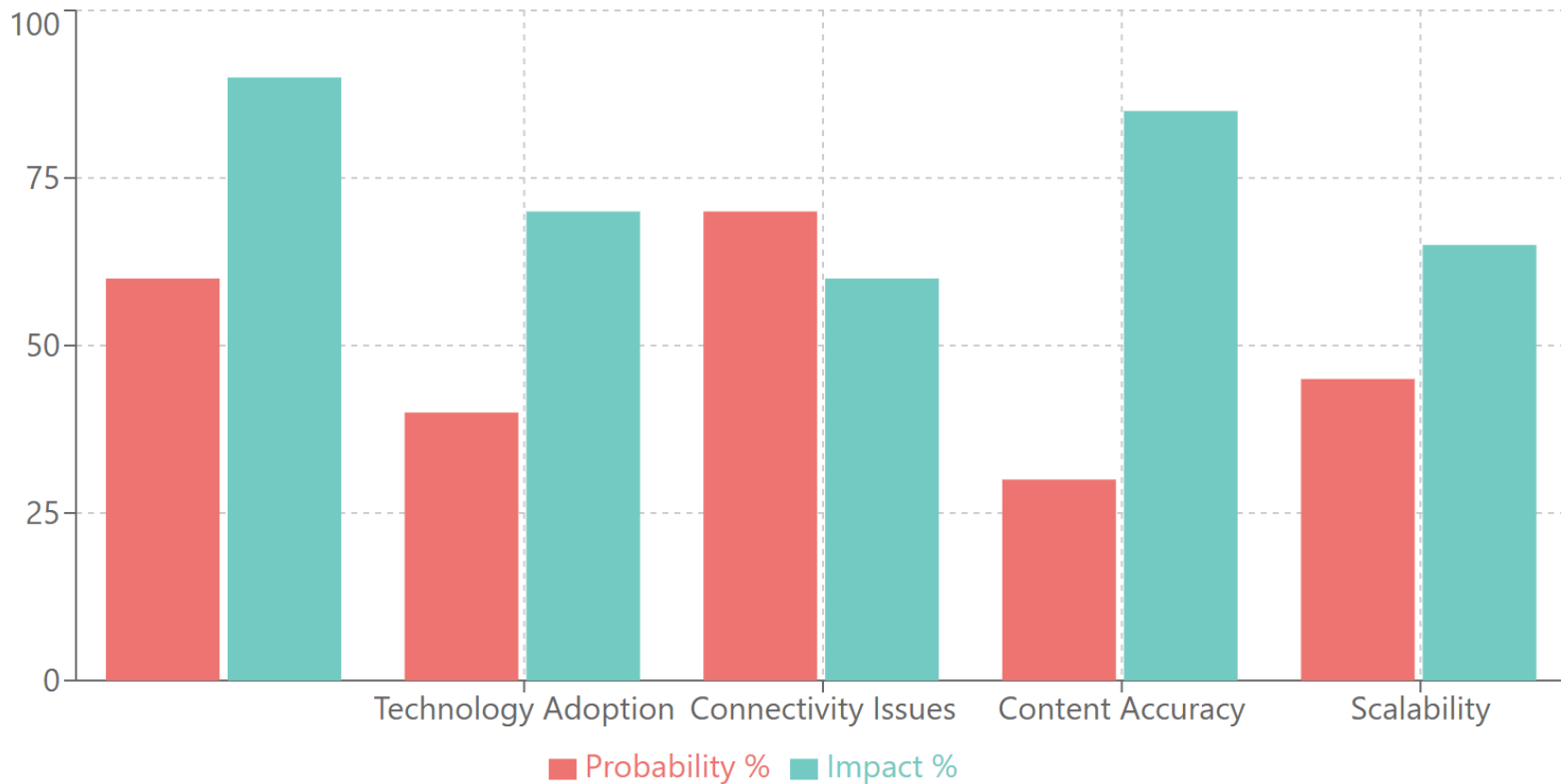
## Medium Priority: Technology Adoption

Community engagement programs, extensive training, user-friendly design

## Medium Priority: Scalability Challenges

Cloud-native architecture, microservices design, automated scaling

## Risk Analysis Matrix (Probability vs Impact)



# Competitive Landscape Analysis

| COMPETITOR             | RURAL FOCUS | AI INTEGRATION | OFFLINE SUPPORT | LOCAL LANGUAGES | COMPETITIVE ADVANTAGE    |
|------------------------|-------------|----------------|-----------------|-----------------|--------------------------|
| Aarogya-Saathi         | High        | High           | High            | High            | Rural-first design       |
| Practo                 | Low         | Medium         | Low             | Medium          | Urban market presence    |
| Apollo TeleHealth      | Low         | Medium         | Low             | Low             | Premium healthcare brand |
| Government Initiatives | High        | Low            | Medium          | High            | Government backing       |

# 5-Year User Adoption Projection

